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30 June 1971
SUPERSEDING
MIL-N-83259 (USAF)
3 April 1970

MILITARY SPECIFICATION

NOISE SUPPRESSOR SYSTEM, AIRCRAFT GROUND RUN-UP A/F32A-19;
FOR A-7D AIRCRAFT

1. SCOPE

1.1 This specification covers one type of demountable, aircraft ground run-up noise suppressor system, designated A/F32A-19.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-N-83155 Noise Suppressor System, Aircraft Turbine Engine Ground Run-Up, General Specification for

STANDARD

Military

MS51336 Lunette - Coupler, Drawbar, Ring

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components. The noise suppressor system shall consist of the following major components:

Item or System Name	See Requirement
Exhaust muffler and augmenter	3.6
Secondary air intake enclosure	3.7

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Item or System Name	See Requirement
Primary air intake muffler	3.8
Aircraft guide tracks	3.9
Aircraft restraint system	3.10.

3.2 General specification. The requirements of MIL-N-83155 apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.3 Design. The noise suppressor system shall be so designed that it will constitute a complete and operable facility capable of satisfying the operational requirements described herein. Except for the concrete foundation (see 3.13), the noise suppressor system supplier shall furnish all components necessary to provide a complete functional system. Anchor bolts will not be provided in the concrete pad; therefore, the system supplier shall install inserts in the concrete or provide other means of bolting the system to the concrete pad. The noise suppressor shall be designed to utilize the foundations and facilities shown on figure 1.

3.3.1 Effect on aircraft structure. The noise suppressor system shall be designed to direct and control primary, secondary, and exhaust flow so as to eliminate any possibility of damage to aircraft surfaces due to buffeting or overtemperature.

3.3.2 Attachment and quick release. The noise suppressor system shall be so designed that after an A-7D aircraft is located in the test position, all components of the system can be engaged and the system ready for operation within a 10-minute period, using a three-man crew. All components of the noise suppressor system that are attached to, or enclose, parts of the aircraft shall also incorporate a quick-release method of detachment. It shall be possible to release all components of the system, including retraction of the primary air intake muffler, and prepare the aircraft for removal from the test position within a 3-minute period, using a three-man crew.

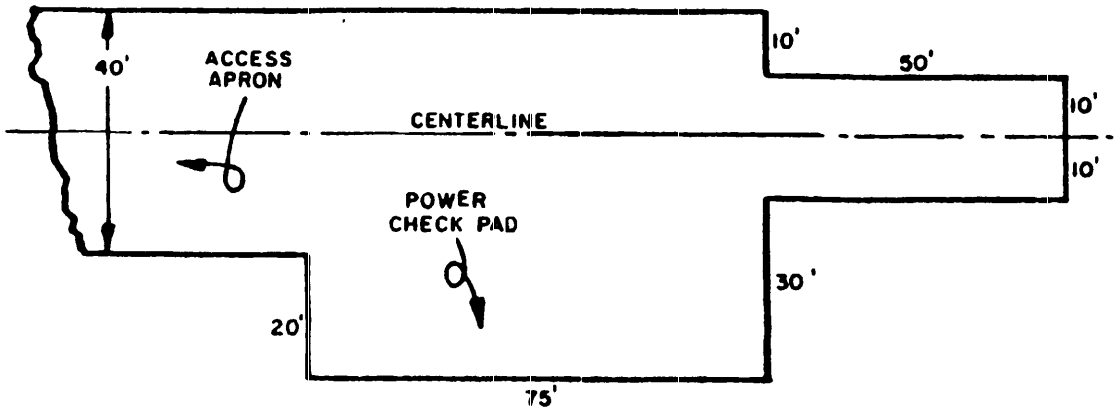
3.3.3 Tolerance of aircraft movement. The noise suppressor system shall accommodate vertical movement of the aircraft resulting from (see 3.10.1):

a. Engine power setting changes from idle to maximum power or any intermediate power level

b. Varying aircraft configurations during ground run-up operations

c. Varying fuel loads and undercarriage oleo strut inflations.

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BASE REQUIREMENTS (Government Furnished)

1. Existing construction: Locate on portland cement pavement (PCP) conforming to light load airfield pavement design, minimum. Slope of pad shall not exceed 1 percent in any direction.
2. New construction: Construct PCP conforming to light load airfield pavement design, minimum, for all but a 20- by 50-foot section. Construct the 20- by 50-foot section to withstand a downward vertical load of 3,000 psf. Minimum thickness of the 20- by 50-foot section shall be 8 inches to provide adequate foundation for insert embedment. Longitudinal centerline of entire pad shall be level with 1/2 percent transverse slope to each side of the centerline.
3. Electrical grounds: Install as required.
4. Floodlights: Install as required.

EQUIPMENT REQUIREMENTS (Contractor Furnished)

1. Primary air intake muffler: Furnish and install guide rails. Embed as necessary to permit passage of the aircraft wheels over the rails (see 3.8).
2. Anchorage: Furnish and install inserts as necessary to secure the noise suppressor equipment (see 3.3).
3. Storm drainage: Provide as required.

*FIGURE 1. Noise Suppressor Power Check Pad for A-7D Aircraft

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3.3.4 Aircraft clearance. Except for a soft or inflatable seal, no part of the noise suppressor system shall make physical contact with the aircraft.

3.3.5 Aircraft control surface movement. Full angular movement of the aircraft horizontal stabilizer and rudder shall be permitted while the aircraft is being positioned in, operated with, or removed from the noise suppressor system. A minimum clearance of 2 inches shall be maintained between the stabilizer and rudder and the noise suppressor system (excluding the soft or inflatable seal) under all conditions.

3.3.6 Aerodynamic and thermodynamic limitations. The gas stream flow velocities and temperatures listed in table I shall not be exceeded in the design of the noise suppressor system components. The maximum allowable flow velocities stated are for smooth, diffuse gas flows at grazing incidence to the facing sheet of the panel.

3.4 Construction. The noise suppressor system shall be demountable as defined in MIL-N-83155. Table I illustrates typical component construction that may be used in the noise suppressor system. The component minimum metal thickness requirements are based on the use of mild steel and have been established as the minimums acceptable to satisfy construction requirements only and in many cases may have to be increased to permit the required acoustical performance of particular components to be achieved. All acoustical panels shall be provided with separators, reducing the height of unsupported acoustical material to 24 to 30 inches, to prevent settling of the acoustical fill materials. Where material or overall thicknesses and acoustical fill densities are not indicated in table I, these values shall be determined by the supplier to satisfy the overall requirements for the noise suppressor system.

3.5 Performance

3.5.1 Acoustical performance

3.5.1.1 Far-field acoustical performance. The noise suppressor system in operation with the A-7D aircraft shall provide grade II acoustical performance as specified in MIL-N-83155 when the sound pressure levels measured 250 feet from the unsuppressed aircraft have values up to and including those specified in table II.

3.5.1.2 Near-field acoustical performance. The noise suppressor system shall satisfy the near-field acoustical requirements of MIL-N-83155 at the positions identified on figure 2.

3.5.1.3 Consideration of possible error in measuring far-field acoustical data. To account for possible instrumentation error, ground reflection, temperature gradients, and other possible errors in recording sound pressure

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TABLE I. Component Aerodynamic, Thermodynamic, and Construction Criteria

COMPONENT	ACCEPTABLE TYPES OF CONSTRUCTION	MAX. FLOW VELOCITY (Ft/Sec)	MAX. MIXED GAS TEMP. (°F)
Exhaust Plenum and Augmenter Walls *	5/16" Solid Inner Shell, Firebox Quality Steel or Equal Reinforced With Structural Steel Members 11 Gage External Cover 4" Thick 6 pcf Acoustical Fill		
Exhaust Silencer Shell	*5/16 inch solid firebox quality steel or equal. If acoustical treatment is provided on the inner surface of the shell, the construction shall be equivalent to that shown below for the exhaust silencer dissipative noise control elements. If lagging is provided on the outer surface of the shell, the external cover shall be 11 gage.		
Exhaust Silencer Dissipative Noise Control Elements	3/16" Perforated Facing S.S. Wire Screen Glass Fiber Cloth Acoustical Fill	.180	600
	3/16" Perf. Facing Porous Metal Seperator 14 Gage Perf. Facing S.S. Wire Screen Glass Fiber Cloth Acoustical Fill	250	700
Exhaust Muffler Resonators and Deflectors	1/4" Firebox Quality Steel or Equal	300	800
Secondary Air Intake Enclosure Walls and Ceilings	16 Gage Perf Facing Glass Fiber Cloth 3/16" External Cover 8" Thick 4 pcf Acoustical Fill	90	
Secondary Air Intake Silencer Shells	16 Gage Perf. Facing Glass Fiber Cloth 3/16" External Cover 4" Thick Acoustical Fill	90	
Secondary Air Intake Silencer Dissipative Noise Control Elements	16 Gage Perf. Facing Glass Fiber Cloth Acoustical Fill	90	
Primary Air Intake Muffler Panels	18 Gage Perf. Facing Glass Fiber Cloth 16 Gage Solid Outer Sheet 4 pcf Acoustical Fill	90	

*NOTE: If augmenter exterior walls are not exposed to the atmosphere, 5/16 inch solid firebox quality steel, or equal, will be considered acceptable construction.

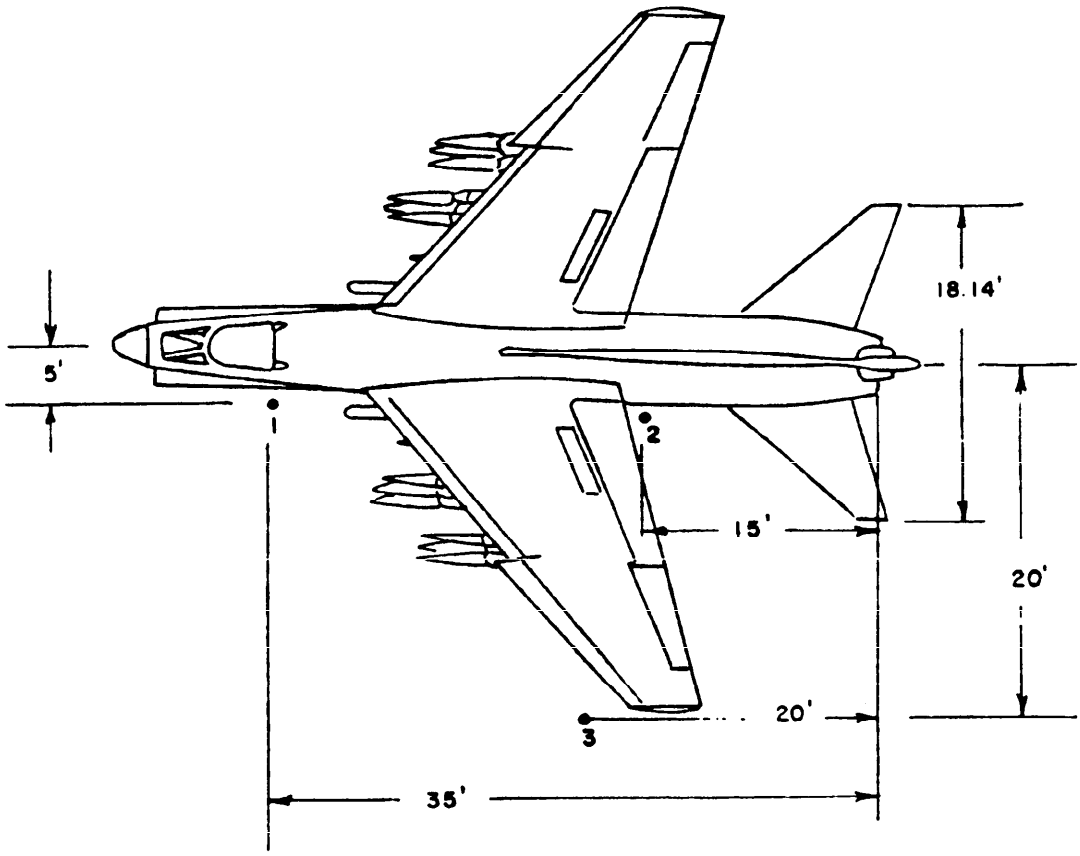
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levels during the far-field acoustical performance evaluation, the noise suppressor system will be considered to have satisfied grade II performance if the measured levels do not exceed grade II levels by more than 4 dB. If an unsuppressed far-field acoustical evaluation is performed under the conditions identified in 3.5.1.4, the measured sound pressure levels will be considered as not exceeding table II values if they are no more than 4 dB higher.

TABLE II. A-7D Maximum Unsuppressed Noise Levels With The TF41 Engine Operating at Full Military Power

POSITION RELATIVE TO ENGINE EXHAUST NOZZLE		SOUND PRESSURE LEVEL (In dB Re 0.0002 DYNES / CM^2)							
ANGLE IN DEGREES (0° FORWARD)	DISTANCE IN FEET	OCTAVE BAND CENTER FREQUENCIES (Hz)							
		63	125	250	500	1000	2000	4000	8000
0	250	95	101	106	111	106	99	95	92
10	250	95	101	109	116	111	107	102	100
20	250	95	104	109	119	115	110	106	105
30	250	96	102	107	118	116	112	107	102
40	250	96	101	106	116	112	110	110	109
50	250	97	101	105	112	111	111	110	107
60	250	99	103	105	112	111	112	112	113
70	250	99	106	105	109	111	111	112	113
80	250	101	106	106	108	112	111	105	112
90	250	101	108	107	109	110	109	110	115
100	250	102	109	109	112	112	115	114	115
110	250	106	112	113	117	116	115	116	120
120	250	112	121	120	122	120	119	118	118
130	250	118	128	126	126	122	121	121	123
140	250	120	129	128	129	127	123	122	126
150	250	119	118	118	111	107	104	105	102
160	250	118	117	116	108	105	102	102	102

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Note: All measurements made with the microphone positioned 5 feet above the ground plane.

FIGURE 2. Near-Field Measurement Positions

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3.5.1.4 Unsuppressed acoustical survey. If grade II acoustical performance, after consideration is given to possible error in measurement as specified in 3.5.1.3, is not provided by the noise suppressor system at one or more measurement points, an unsuppressed acoustical survey may be performed. This unsuppressed survey shall be performed, in accordance with contractual conditions identified by the procuring activity, if the supplier considers the failure to be caused by the engine used in the test producing a higher sound pressure level at the measurement point(s) in question than the table II value(s) and requests that the unsuppressed survey be performed. The surface overwhich the unsuppressed survey is conducted shall be as hard or harder than the surface overwhich the suppressed survey was performed. Corrections for molecular absorption as specified in MIL-N-83155 shall be applied to the unsuppressed data. The noise suppressor system shall be considered acoustically acceptable if grade II levels are not exceeded by a greater amount than the measured unsuppressed levels exceed the values in table II after both measured suppressed and unsuppressed data are given the consideration for possible error in measurement as specified in 3.5.1.3.

3.5.2 Effect on engine performance. The engine performance parameters to be used in determining the effect of the noise suppressor system on the engine shall be those parameters recorded during engine trimming operations as follows:

N_{HP}	-	High pressure rotor speed	(rpm)
N_{LP}	-	Low pressure rotor speed	(rpm)
P_{t5}	-	Low pressure turbine discharge total pressure	(in. Hg)
T_{t5}	-	Low pressure turbine discharge total temperature	(^{o}C)
w_f	-	Fuel flow	(lb/hr)
T_{am}	-	Ambient static temperature	(^{o}F)
P_{am}	-	Ambient static pressure	(in. Hg).

3.5.3 Operational capability. The noise suppressor system shall be capable of operating under the conditions specified in MIL-N-83155 with the A-7D aircraft (TF41-A-1 turbofan engine).

3.6 Exhaust muffler and augmenter. The selection of construction materials and gages for the exhaust muffler and augmenter shall be determined by the design life and operational capability requirements of MIL-N-83155 and the requirements herein. The exhaust muffler and augmenter shall be cooled by augmented secondary air during operation of the TF41 turbofan engine at all

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power settings. Water cooling will not be permitted. All components, such as flow diffusers, resonators, deflectors, or dissipative noise control elements, located within the exhaust muffler or augmenter shall be designed to permit replacement without dismantling the basic muffler or augmenter structure.

3.6.1 Augmenter. The augmenter shall collect the engine exhaust gas stream and augmented cooling air and shall provide an enclosed path for these gases from the secondary air intake enclosure to the exhaust muffler. The augmenter design shall take into account the range of engine exhaust nozzle centerline positions resulting from changes in engine power setting, varying fuel loads, and undercarriage oleo strut inflation (see 3.10.1).

3.6.2 Exhaust muffler. The exhaust muffler shall provide an acoustically treated path for the exhaust gases exiting from the augmenter. The final direction of the exhaust gases exiting from the muffler stack shall be vertical. At no position within the exhaust muffler shall the exhaust gas stream be permitted to impinge, either directly or indirectly, on the concrete foundation. Provisions for surface drainage shall be made to prevent the accumulation of water within the exhaust muffler. If it is not possible for a man to easily enter the exhaust muffler through the augmenter, a personnel access panel shall be provided in one wall of the muffler to
* permit interior inspection and maintenance. The panel shall be hinged at the side and shall be accessible from ground level. The panel shall not affect the acoustical performance of the exhaust muffler.

3.7 Secondary air intake enclosure. The secondary air intake enclosure shall provide an acoustically treated path between the aircraft and the augmenter for the turbofan engine exhaust gases and the augmented air. The enclosure shall be of sufficient size to permit the augmentation of the proper amount of secondary air required to cool the exhaust muffler and augmenter, as determined to comply with the design life requirements of this specification. All components, such as deflectors or dissipative noise control elements, located within the enclosure shall be designed to permit replacement without dismantling the basic enclosure structure. Provisions for surface drainage shall be made to prevent the accumulation of water within the enclosure.

3.7.1 Acoustical seal. An acoustical seal shall be provided in the forward wall of the enclosure and adjacent to the aft fuselage of the aircraft to prevent the forward radiation of noise out of the enclosure. The seal shall
* consist of a soft, nonflammable, compressible material capable of withstanding the operational environment and of preventing damage to the aircraft. The seal shall be adjustable to allow enlarging the opening for positioning the aircraft with the noise suppressor system. The seal shall be mounted in a floating panel or otherwise designed to accommodate vertical aircraft movements resulting from changes in engine power setting, varying fuel loads, and undercarriage oleo strut inflation (see 3.10.1).

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3.7.2 Pressure loss. The pressure loss caused by the secondary air intake enclosure shall not exceed 2 inches of water.

3.7.3 Access doors. Two personnel access doors (one on each side) shall be incorporated in the secondary air intake enclosure. Each door shall be 80 inches high by 36 inches wide and shall be equipped with an 18- by 24-inch double-pane, shatterproof glass observation window located with the 24-inch dimension horizontal and centered 66 inches above ground level. The doors shall be so constructed that they are readily accessible, unobstructed, and quick opening. The doors shall open outward. The door latches shall open from the inside or outside by a single motion of the hand. A 6-inch high by 10-inch wide panel shall be incorporated in each door to provide access for standard ground fire-fighting equipment. The horizontal centerline of the panels shall be 45 inches above ground level. The panels shall be hinged along the bottom edge and open outward. Each panel shall be held in the closed position by a latch that can be quickly released. A handle shall be provided on each panel to permit the panel to be pulled open while a negative pressure caused by operation of the noise suppressor system exists within the enclosure. The panels shall be marked ACCESS FOR FIRE EXTINGUISHER.

* 3.8 Primary air intake muffler. The primary air intake muffler shall consist of a single, rail-mounted unit that can be manually positioned by not more than two men. The muffler shall assist in the overall acoustical treatment of the noise suppressor system.

* 3.8.1 The muffler shall present a smooth aerodynamic path for the airflow to prevent turbulence at the aircraft air inlet. The pressure drop through the muffler at any position shall not exceed 2 inches of water. Provisions shall be included to prevent the collection of debris, water, et cetera, in the muffler and the ingestion of ground maintenance personnel and foreign object damage into the muffler.

* 3.8.2 Positioning provisions. The primary air intake muffler shall be designed for positioning from the left-hand side of the aircraft on the rails described in 3.8.8. The rails shall have stops incorporated to assist in the correct positioning of the muffler. A hinged or sliding section shall be incorporated in the aft right-hand wall of the muffler to provide clearance for the radome, inlet screen, and nose section of the aircraft when the muffler is being moved into position. When the movable section is closed, it shall provide the same degree of acoustical performance as the other walls of the muffler. Positive latches shall be provided to lock the movable section in both the open and closed positions.

* 3.8.3 Towbar. A hinged-type towbar shall be provided on the right-hand side of the muffler to permit a tow vehicle to assist in positioning the muffler under adverse wind and weather conditions. The towbar shall be so located that

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it will be not less than 4-1/2 feet forward of the aircraft radome when in its normal operational position perpendicular to the aircraft longitudinal centerline. The towbar shall be hinged and pivoted so as to move in both the horizontal and vertical directions. A positive-type latch shall be provided to hold the towbar in the up (vertical) position at any horizontal angle. The towbar shall also have a stop in the up position to prevent contact with the wall of the muffler. The towbar shall have a lunette eye conforming to MS51336.

* 3.8.4 The muffler shall remain in the correct operational position in relation to the aircraft, once having been so positioned, without any physical attachment between the aircraft and muffler. A soft, compressible seal may be used between the muffler and aircraft contours to prevent damaging the aircraft during positioning of the muffler and while the aircraft is being operated.

* 3.8.5 The primary air intake muffler shall be designed to allow easy and unobstructed access to, and exit from, the aircraft cockpit. If the muffler interferes with the standard method of access to and exit from the cockpit, an alternate method shall be provided as an integral part of the muffler. A clear line-of-sight shall be maintained between the cockpit and ground maintenance personnel.

* 3.8.6 Vertical aircraft movements. The primary air intake muffler shall accommodate vertical aircraft movements resulting from changes in engine power setting, varying fuel loads, and undercarriage oleo strut inflation (see 3.10.1).

* 3.8.7 Restraining provisions. The muffler shall be provided with some means of restraint, when in the stored position, to prevent moving under steady wind conditions of 60 mph with gusts up to 75 mph. When so restrained in the stored position with the towbar latched in the up position, 25 feet of clearance shall be maintained between the intake muffler and the longitudinal centerline of the aircraft.

* 3.8.8 Intake muffler rails. The rails shall be located perpendicular to the longitudinal centerline of the aircraft and shall be level over their entire length within 1/8 inch in 10 feet. The tops of the rails shall be not more than 1 inch above the concrete pad at the longitudinal centerline of the aircraft. The concrete pad may be cut to a maximum depth of 2-1/2 inches in order to embed a portion of the rails. Ramps shall be provided on each side of the rails at all positions where the tops of the rails are more than 1 inch above the surface of the concrete pad, with the exception of the stored position of the intake muffler at the left-hand end of the rails. The inclination of the ramps shall be not greater than 10 percent.

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3.9 Aircraft guide tracks. An aircraft undercarriage guide track system shall be provided to facilitate accurate and easy positioning of the aircraft with respect to the exhaust muffler and secondary air intake enclosure.

* 3.10 Aircraft restraint system. Restraining equipment shall be employed to absorb the thrust developed during run-up of the A-7D aircraft in the noise suppressor system. The aircraft shall be restrained by means of chocks, fore and aft of the main landing gear. The chocks shall be fastened to the main landing gear guide track.

* 3.10.1 Aircraft restraint system to reduce vertical movement. If approved by the airframe manufacturer and the procuring activity (see 6.2), a restraint system may be used to reduce the vertical movement of the aircraft.

* 3.11 Engine vibration levels. The overall vibration levels associated with the operation of the installed TF41 turbofan engine in the A-7D aircraft in the noise suppressor system shall not exceed the maximum allowable vibration limits permitted by applicable engine technical orders.

3.12 Finishes and protective coatings. Cleaning, painting, and plating shall be as specified in MIL-N-83155. Unless specified otherwise by the procuring activity (see 6.2), the finish colors of the various components of the noise suppressor system shall be as follows:

Primary air intake muffler:

Exterior - Gloss yellow, film designation DG

Interior - Unpainted, galvanized surface finish in accordance with finish No. P-116

Secondary air intake enclosure:

Exterior - Gloss yellow, film designation DG

Interior - Gloss yellow, film designation DG or unpainted, galvanized surface finish in accordance with finish No. P-116

Exhaust muffler and augmenter:

Exterior - Lusterless black, film designation UL

Interior - High temperature aluminum paint (two coats)

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Aircraft guide tracks:

Gloss yellow, film designation DG

Final painting shall be done after the system is assembled.

* 3.13 Government-furnished property. A concrete power check pad as shown on figure 1 will be furnished by the Government at each site (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. The inspection and testing of the noise suppressor system shall be in accordance with MIL-N-83155 and as specified herein.

* 4.2 Preproduction tests. The preproduction tests, to be performed on the first noise suppressor system delivered and installed, shall consist of all tests specified in MIL-N-83155 and all tests under 4.4 (see 6.2).

4.3 Acceptance tests. The acceptance tests, to be performed on all noise suppressor systems delivered subsequent to the first system, shall consist of the individual test specified in MIL-N-83155 and the tests specified in 4.4.5, 4.4.6, 4.4.7, and 4.4.10 herein.

4.4 Test methods

4.4.1 Acoustical evaluation. The acoustical performance of the noise suppressor system shall be evaluated as specified in MIL-N-83155 with the A-7D aircraft.

4.4.2 Effect on engine performance. The effect on engine performance test specified in MIL-N-83155 shall be performed with the A-7D aircraft. The engine performance parameters listed in 3.5.2 shall be recorded for the following:

Run-up configurations

a. Unsuppressed aircraft

b. Aircraft positioned in the complete noise suppressor system

Modes of aircraft operation

* a. Military (see 6.3.2)

b. Maximum continuous

c. 90 percent maximum continuous

d. 75 percent maximum continuous.

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4.4.2.1 Correlation data. Any difference in the data for the measured performance parameters between the two configurations shall be documented on correlation graphs developed by the noise suppressor system supplier.

4.4.3 Engine vibration level. During the tests conducted under 4.4.2, the engine vibration levels for the run-up of the aircraft engine shall be recorded for the positions stated in the applicable engine technical orders. The maximum allowable vibration limits specified in 3.11 shall not be exceeded at any position in order for the noise suppressor system to pass this test. (See 6.2.)

* 4.4.4 Durability. The durability test specified in MIL-N-83155 shall not be applicable to this equipment. The operation of the A-7D aircraft with the noise suppressor system for the acoustical test and for the effect on engine performance test will be considered as satisfying the durability test requirements.

4.4.5 Attachment and quick release. The methods of installing and quickly releasing the aircraft in the noise suppressor system shall be tested to determine compliance with 3.3.2.

4.4.6 Tolerance of aircraft movement. The noise suppressor system shall be checked to insure that any vertical aircraft movement resulting from changes in engine power setting, varying fuel loads, and undercarriage oleo strut inflation can be accommodated without damage to the aircraft structure while permitting satisfactory operation of the noise suppressor system.

4.4.7 Control surface movement. The noise suppressor system shall be checked to insure that full angular movement of the horizontal stabilizer and the rudder can be obtained while the aircraft is positioned in the noise suppressor system without causing any physical interference between the aircraft or suppressor.

4.4.8 Pressure loss in primary and secondary air intakes. The noise suppressor system supplier shall demonstrate that the pressure loss in the primary and secondary air intakes does not exceed 2 inches of water. The pressure differences shall be measured with a U-tube water manometer with an accuracy of $\pm$0.1 inch of water. For the secondary air intake, measurements shall be taken at three positions within the enclosure, to be approved by the procuring activity (see 6.2). For the primary air intake muffler, measurements shall be taken at the following positions:

a. Directly ahead of the primary aircraft air inlet on the inlet centerline and 6 inches from the muffler wall

b. In line with the primary aircraft air inlet at the inlet centerline height and 6 inches from the muffler sidewall

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c. Directly below the primary aircraft air inlet and 6 inches above the muffler floor.

4.4.9 Gas stream velocities and temperatures. Tests shall be conducted to verify that the gas stream velocities and temperatures specified in table I are not exceeded.

4.4.10 Mobility of primary air intake muffler. The primary air intake muffler shall be tested to insure that movement and positioning can be accomplished by two men without strain. The force required to move the muffler on level rails shall be measured and shall not exceed 100 pounds. The average surface wind shall not exceed 5 knots while this measurement is being performed. The muffler shall also be positioned using a tow vehicle to demonstrate the correct operation of the towbar.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The A/F32A-19 noise suppressor system is intended for use in reducing the noise generated during ground maintenance run-up of the A-7D aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Procedures for approval of the design of vertical movement restraining method (see 3.10.1)

c. If colors used in painting suppressor components for selected sites are to be different from those specified in 3.12

d. Location and conditions for preproduction testing (see 4.2) including:

(1) Extent of contractor support and participation during testing (see 4.2)

(2) Extent of Government participation in testing (see 4.4.3)

(3) Procedure for approval of test positions (see 4.4.8).

* e. Description of the government-furnished foundation and facilities at the installation site (see 3.13).

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6.3 Definitions. For the purpose of this specification, the following definitions shall apply:

6.3.1 Stable engine operation. The TF41 turbofan engine is considered to be stable in operation when the engine parameters N_{HP}, N_{LP}, P_{t5}, and T_{t5} (see 3.5.2) can be maintained within the following limits:

N_{HP}	±0.1 percent of reading
N_{LP}	±0.1 percent of reading
P_{t5}	±0.25 inch Hg
T_{t5}	±1°C.

* 6.3.2 Military power. Military power shall be intermediate power as specified in the applicable TF41 turbofan engine technical order.

6.4 Reference material. AFM 86-4 Standard Facility Requirements may be used as a source of information in connection with this specification.

* 6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4920-F278

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